The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for Social Sciences

Tutorial Sheet 3- Functions **April, 2023.**

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1. For $U = \{0, 1, 2, 3, 4\}, A = \{0, 4\}, B = \{2, 3\}$, find

   (a) $A \times B$ (b) $A' \times (A - B)$

2. For $A = \{a, b, c, d, e\}, B = \{a, e, o, i, u\}, C = \{1, 2, 4, 8\}$ find the following

   (a) $(A - B) \times (B - A)$ (b) $(A \times C) \cap (B \times D)$

3. A relation R is defined in $\mathbf{Z}^+$ by $R = \{(a, b)|a^3 + b^2 < 20\}$. List all elements of R, and determine its range and domain.

4. Let A and B be two sets such that $A = \{2, 5, 7, 8, 10, 13\}$ and $B = \{1, 2, 3, 4, 5\}$. Suppose R is defined by $R = \{(x, y) : x = 4y - 3, x \in A, y \in B\}$.

   (a) List the elements of the set R .

   (b) Display the relation R on the arrow diagram.

   (c) What type of relation is this?

   (d) Find the domain, range and co-domain of R.

5. Find the inverse of the relation $R = \{(4, 8), (-3, -6), (7.1, 2.3)\}$.

6. In each of the following relations, illustrate the information given on an arrow diagram and state which of these relations are functions.

   (a) $\{(0, -1), (2, 2), (1, -2), (3, 0), (1, 1)\}$

   (b) $\{(1, 4), (3, 4), (7, 3)\}$

   (c) $\{(4, 3), (4, 7), (3, 4)\}$

   (d) $\{(1, 2), (1, 3), (2, 3)\}$

   (e) $\{(1, 2), (2, 3), (3, 4)\}$

7. Prove that the following functions are one-to-one

(a) $f(x) = -3x^3 - 1$

(b) $f(x) = \sqrt{x}$

8. If the function f is defined by

$$f(x) := \begin{cases} -3x+2 & \text{for } x < 0, \\ x^2+1 & \text{for } 0 \le x \le 4, \\ -1 & \text{for } x > 4, \end{cases}$$

compute $f(0)$, $f(7)$, $f(-2)$.

9. For each of the following functions, find the expression for $\frac{f(x+h)-f(x)}{h}$

(a) $f(x) = 4x + 5$

(b) $f(x) = x^2 - 3x$

(c) $f(x) = \frac{1}{x}$

(d) $f(x) = \frac{1}{\sqrt{x}}$

10. For each of the following functions, find/state the domain

(a) $f(x) = x - 5$

(b) $f(x) = \frac{1}{x+1}$

(c) $f(x) = \sqrt{x+4}$

(d) $f(x) = \sqrt{x^2+4}$

11. Let f and g be two functions. Find $(fog)(x)$ and $(gof)(x)$, specifying the domain in each case:

(a) $f(x) = 3x + 4,\ \ g(x) = x^2 + 1$

(b) $f(x) = x^2 - x - 1,\ \ g(x) = x + 4$

(c) $f(x) = \sqrt{x-2},\ \ g(x) = 3x - 1$

(d) $f(x) = \frac{1}{x-1},\ \ g(x) = \frac{2}{x}$

12. If $f(x) = x^2 - 2,\ \ g(x) = x + 4$, find $(fog)(4)$ and $(gof)(4)$.

13. Determine which of the following functions are even, odd or neither odd nor even.

(a) $f(x) = x^2 - 4x + 5$

(b) $f(x) = 2x + 1$

(c) $f(x) = x^2 + 4x - 10$

(d) $f(x) = \sqrt[3]{x}$

(e) $f(x) = 5 - 5x - 3x^2$

(f) $f(x) = 3x^4 + x^2 - 7$

(g) $f(x) = \frac{x^2+4}{x^3-x}$

(h) $f(x) = \frac{3}{x^2+2}$

14. For each of the following functions, (i) list its domain and range, (ii) form the inverse function f^{-1} and list the domain and range of f^{-1}.

(a) $f = \{(1,5),(2,9),(5,21)\}$

(b) $f = \{(0,0),(2,8),(-1,-1),(-2,-8)\}$

15. Show that the following functions are inverses of each other.

(a) $f(x) = 5x - 9,\ \ g(x) = \frac{x+9}{5}$

(b) $f(x) = \frac{1}{x-1}$ for $x \neq 1$, $g(x) = \frac{x+1}{x}$ for $x \neq 0$.

16. If $f(x) = 2x + 3$ and $g(x) = 3x - 5$, find

(a) $(fog)^{-1}(x)$

(b) $(f^{-1}og^{-1})(x)$

(c) $(g^{-1}og^{-1})(x)$

(d) $(fog)^{-1}(2)$

(e) $(f^{-1}og^{-1})(4)$

(f) $(g^{-1}og^{-1})(-10)$

17. If $f(x) = x - 4$ and $g(x) = \dfrac{3}{x+1}$, solve the equation $(f \circ g)^{-1}(x) = 2$.

18. The weekly cost C of producing x uinits in a manufacturing process is given by the function $C(x) = 70x + 800$. The number of x units produced in t hours is given by $x(t) = 20t$. find the cost of producing x units in 25 minutes.